

- Q.22 With a neat diagram, explain aircraft communication system.
- Q.23 Describe briefly, what do you mean by Avionics packaging?
- Q.24 Demonstrate the working of electronic Flight system
- Q.25 How is Data bus integrated in Business Jets?
- Q.26 How does circuit controller work?
- Q.27 What is the role of avionics equipment fit?
- Q.28 How are Line Replaceable Units used?
- Q.29 Describe a stable navigation system?
- Q.30 What are the differences between Transmitter and Receiver?
- Q.31 How do you find Frequency response of FBW system?
- Q.32 What is dead reckoning navigation system and explain any one in detail.
- Q.33 Describe the term failure revival.
- Q.34 Explain DVI?
- Q.35 Explain electronic warfare.

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Explain the nature of major microelectronic devices with examples.
- Q.37 Draw and explain the function of HDD, HUD and MFD.
- Q.38 What do you mean by redundancy and failure survival? How both are connected? What are common modes on failures?

No. of Printed Pages : 4
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6th Sem / AME Subject:- Aircraft Avionics

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 What filter is used by the FMS to estimate the position of the aircraft from various sensors?
- Kalman filter
 - Position error filter
 - GPS filter
 - Gyro and accelerometer filter
- Q.2 What does the outer loop of an autopilot do?
- Generate commands to the inner flight control loop
 - Access the flight control surfaces to control the pitch and roll of the aircraft
 - Access the flight control surfaces to control the altitude and heading of the aircraft
 - Generate commands for the pilot to follow
- Q.3 The main reason for using a collimated display in HUD is _____
- More information can be displayed
 - Wide view of display
 - Information displayed is focused at infinity
 - Fast screen refresh rates

(20)

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187763/147763

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187763/147763

- Q.4 Which of the following is the correct order for the path of light in a HUD?
- CRT-> mirror-> collimating lens-> combiner glass-> pilot
 - CRT-> collimating lens-> mirror-> combiner glass-> pilot
 - CRT-> mirror-> combiner glass-> collimating lens-> pilot
 - Collimating lens-> CRT-> mirror-> combiner glass-> pilot
- Q.5 Which of the following is not a navigation mode used by FMS for computing position of the aircraft?
- Inertial (IRS) - GPS
 - Inertial (IRS) - DME/DME
 - Inertial (IRS) - VOR/DME
 - GPS only
- Q.6 The transmission of data by dots and dashes is an example of _____
- Amplitude modulation
 - Frequency modulation
 - Continuous wave transmission
 - Discrete signal transmission
- Q.7 What is the interval for updating data in the navigation database?
- 1 hour
 - 1 month
 - 24 hours
 - 28 days
- Q.8 Which of the following are not controlled by the HUD processor?
- Power supply to HUD
 - Brightness level and contrast level
 - Symbol and font generation
 - Computing flight parameters

- Q.9 Which one of the following is not a navigation mode used by FMS for computing position of the aircraft?
- Inertial (IRS) - GPS
 - Inertial (IRS) - DME/DME
 - Inertial (IRS) - VOR/DME
 - GPS only
- Q.10 _____ is used to provide continuous tuning over the desired range?
- Variable frequency oscillator
 - Frequency synthesizer
 - Variable Frequency producer
 - Frequency multiplier

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Name any one avionic sub system?
- Q.12 What are different memory devices?
- Q.13 What do you mean by Avionics pack?
- Q.14 What is the principle of inertial navigation?
- Q.15 What is Flyby wire system?
- Q.16 What is the use of oscillator?
- Q.17 What is compass swing?
- Q.18 What is HDD?
- Q.19 What is DVI?
- Q.20 What is failure survival?

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Explain one of the most versatile electronic circuit.